Motivation & Context → Data → Analysis Pipeline → Tricky Decision → Bias & Uncertainty → Results → Next Steps

Predicting Short-Term Energy Demands by Weather Patterns

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3/27/2026

Motivation & Research Question

- U.S. electricity demand fluctuates heavily with weather, and grid operators must forecast hours ahead to balance supply
- Accurate short-term forecasts reduce energy waste, lower costs, and prevent blackouts

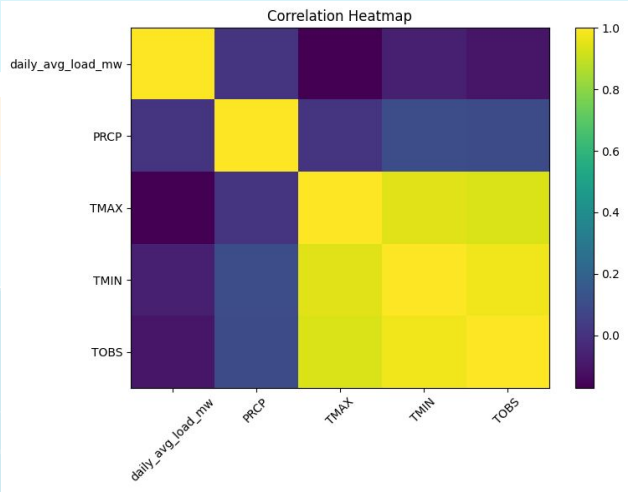
- **Hypothesis:** Adding weather variables (temp, humidity, precipitation) to an ARIMAX model will reduce forecasting error (MAPE & RMSE) vs. a baseline ARIMA

- **Research Question:** To what extent do weather variables improve short-term regional electricity demand forecasts?

- **Initial modeling approach:** Baseline ARIMA vs. ARIMAX with independent weather predictors; evaluated on MAPE and RMSE

Data Acquisition & Explanation

PJM Electricity Data



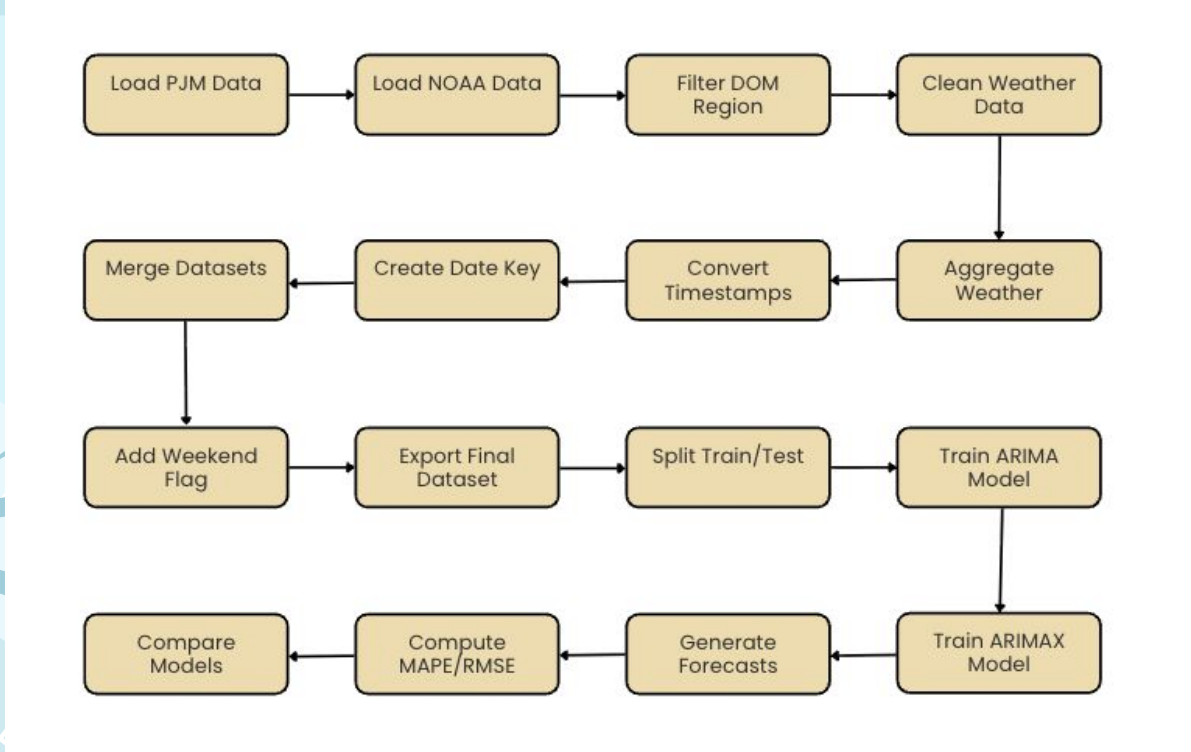
Correlation Heatmap of Forecasting Variables

- PJM Interconnection DataMiner2
- Dominion (DOM) load area
- Hourly metered load in MW
- Jan 2025 – Feb 2026 (8,760+ hourly rows)
- CSV; freely available via PJM portal

NOAA Weather Data

- NOAA Daily Summaries (CDO order)
- 6 Charlottesville-area weather stations
- TMIN, TMAX, PRCP (daily averages)
- Same date range
- CSV; public domain

Analysis Pipeline



Analysis Decision Making

Problem:

- Initially predicted each day's demand using only that day's weather variables and time stamps
- Produced poor predictions
- ARIMA and ARIMAX could not account for daily, weekly, and seasonal patterns simultaneously

Impact:

- Lag structure dramatically improved forecast accuracy over the timestamp and weather-only approach
- Reframed weather's role: a supplementary signal rather than the primary driver

Our Approach:

- Kept ARIMA/ARIMAX for comparison
- Encoded timestamps as hourly, daily, and monthly dummies
- Incorporated lag features by using past demand values as predictors
- Used HistGradientBoostingRegressor instead of ARIMAX to include weekly and seasonal variation via the lag features

Bias & Uncertainty

Geographic

Six stations around Charlottesville may not capture conditions across the full DOM load area (which spans a much larger region)

Seasonal Imbalance

Test set (Jan–Feb 2026) is only winter, performance may not generalize to summer peaks

Station Dropout

Dropping rows with missing TMAX/TMIN/PRCP means some dates are represented by fewer stations, shifting the daily average

Calendar Effects

No focused consideration for special days or periods in the year (e.g. holidays)

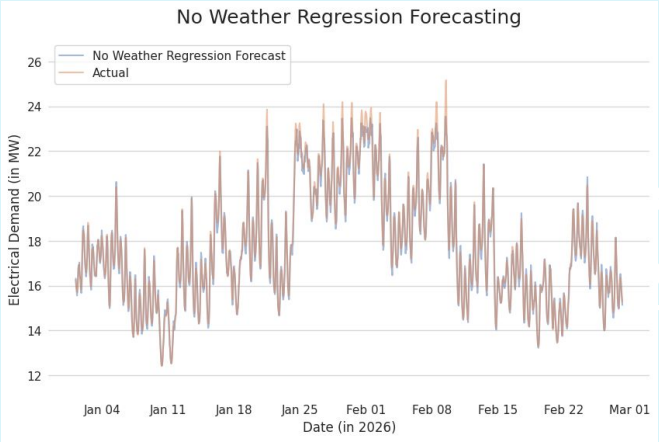
Results & Conclusion

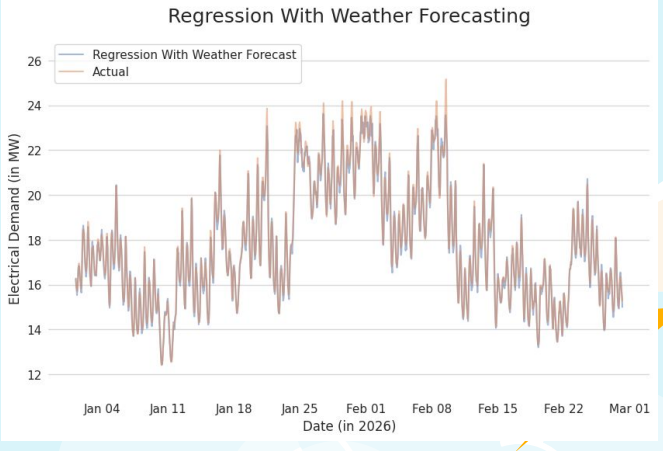
Findings:

- Regression without weather tracked actual demand closely, suggesting lag-based features drove most of the predictive power
- Regression with weather performed similarly, indicating weather added marginal improvement over lag alone
- SARIMAX captured the daily cyclical pattern but over-smoothed peaks and troughs and failed to capture weekly and seasonal cycles

Conclusion:

- **Incorporating weather variables lowered the RMSE and MAE, but only slightly**





Next Steps

Better weather alignment

Extended predictors

Larger test window

References

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[4] PJM Interconnection, “PJM | Data Miner 2 Hourly Load: Metered,” [Online]. Available: https://dataminer2.pjm.com/feed/hrl_load_metered.

[5] NOAA National Oceanic and Atmospheric Administration Home, “National Centers for Environmental Information Data Tools: Find a Station,” [Online]. Available: https://www.ncei.noaa.gov/cdo-web/datatools/findstation.

Questions

WEATHER
climate
heat
air
wind
pressure
temperature
precipitation
humidity
sleet
desertification
Fahrenheit
Celsius
evaporation
satellite
cyclone
cold
blizzard
atmosphere
cumulus
hurricane
gauge
condensation
lightning
altitude
tropical
wave
dew point
cirrus
drought
season
monsoon
elevation
thermometer
barometer
circulation
flood
breeze
tornado
forecast
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